

15.3 The analysis of variance table from projections

Whether the treatments differ at all is answered by the F test, with the analysis of variance table as its bookkeeping. Both come from one orthogonal decomposition of the observation space into the constants, the treatment space and the error space, and every entry of the table is a property of one piece.

15.3.1 Three subspaces

Write $\mathbf{P}_0 = n^{-1}\mathbf{1}\mathbf{1}^\top$ for the projection onto the constants and $\mathbf{M}$ for the projection onto $\mathcal{C}(\mathbf{Z})$ (Eq. 15.1.1). Since $\mathbf{1} \in \mathcal{C}(\mathbf{Z})$, Theorem 6.5.1 applies to the pair $\text{span}(\mathbf{1}) \subseteq \mathcal{C}(\mathbf{Z})$ and gives the orthogonal decomposition

$$\mathbb{R}^n = \underbrace{\text{span}(\mathbf{1})}_{\dim 1} \oplus \underbrace{\mathcal{T}}_{\dim g-1} \oplus \underbrace{\mathcal{C}(\mathbf{Z})^\perp}_{\dim n-g}, \quad \mathbf{I} = \mathbf{P}_0 + (\mathbf{M} - \mathbf{P}_0) + (\mathbf{I} - \mathbf{M}), \quad (15.3.1)$$

with the treatment space $\mathcal{T} = \mathcal{C}(\mathbf{Z}) \cap \mathbf{1}^\perp$ of Section 15.2 and $\mathbf{P}_\mathcal{T} = \mathbf{M} - \mathbf{P}_0$. The three projections act on the data in the familiar way:

$$(\mathbf{P}_0 \mathbf{y})_{kj} = \bar{y}, \quad (\mathbf{P}_\mathcal{T} \mathbf{y})_{kj} = \bar{y}_k - \bar{y}, \quad ((\mathbf{I} - \mathbf{M}) \mathbf{y})_{kj} = y_{kj} - \bar{y}_k.$$

So every observation is split as $y_{kj} = \bar{y} + (\bar{y}_k - \bar{y}) + (y_{kj} - \bar{y}_k)$, grand mean plus treatment deviation plus residual, and the three vectors so formed are mutually orthogonal. Pythagoras gives

$$\sum_{k,j} y_{kj}^2 = n\bar{y}^2 + \sum_k n_k(\bar{y}_k - \bar{y})^2 + \sum_{k,j} (y_{kj} - \bar{y}_k)^2. \quad (15.3.2)$$

The middle term is the **between-groups sum of squares** $\text{SSB} = \|\mathbf{P}_\mathcal{T} \mathbf{y}\|^2$, and the last is the **within-groups** or residual sum of squares $\text{SSE} = \|(\mathbf{I} - \mathbf{M}) \mathbf{y}\|^2$ of Eq. 15.1.2. Their sum $\sum_{k,j} (y_{kj} - \bar{y})^2$ is the corrected total sum of squares. The same decomposition, with the distributions attached, was the first application of Cochran's theorem (Example (The one-way layout)).

15.3.2 The F test

Theorem 15.3.1 — The one-way F test

In the normal one-way model of Definition 15.1.1, with mean vector $\boldsymbol{\theta} = \mathbf{Z}\boldsymbol{\mu}$, put

$$\text{MSB} = \frac{\text{SSB}}{g-1}, \quad s^2 = \text{MSE} = \frac{\text{SSE}}{n-g}, \quad F = \frac{\text{MSB}}{\text{MSE}}.$$

- (a) $\bar{Y}$, $\text{SSB} = \sum_k n_k(\bar{Y}_k - \bar{Y})^2$ and $\text{SSE} = \sum_{k,j} (Y_{kj} - \bar{Y}_k)^2$ are mutually independent.
- (b) $\text{SSE}/\sigma^2 \sim \chi^2(n-g)$, and $\text{SSB}/\sigma^2 \sim \chi^2(g-1, \gamma)$ with

$$\gamma = \frac{\|\mathbf{P}_{\mathcal{T}}\boldsymbol{\theta}\|^2}{\sigma^2} = \frac{1}{\sigma^2} \sum_{k=1}^g n_k (\mu_k - \bar{\mu})^2, \quad \bar{\mu} = \frac{1}{n} \sum_{k=1}^g n_k \mu_k. \quad (15.3.3)$$

- (c) $F \sim F(g-1, n-g, \gamma)$, and $\gamma = 0$ iff $\mu_1 = \dots = \mu_g$. In particular, when the means are equal, $F \sim F(g-1, n-g)$ whatever their common value and σ^2 .
- (d) F is the statistic of the F test of the reduced model $E(\mathbf{Y}) \in \text{span}(\mathbf{1})$ against $\mathcal{C}(\mathbf{Z})$, and of the testable hypothesis $\alpha_1 = \dots = \alpha_g$ in the effects form. The test that rejects when $F > F_\alpha(g-1, n-g)$ is the likelihood ratio test of equal means. Its power is a continuous, strictly increasing function of γ , equal to α at $\gamma = 0$.

Proof. By Eq. 15.3.1 the matrices $\mathbf{P}_0$, $\mathbf{P}_{\mathcal{T}}$ and $\mathbf{I} - \mathbf{M}$ are symmetric, idempotent and mutually orthogonal, with ranks 1, $g-1$ and $n-g$. Theorem 4.4.4 gives (a), since $\bar{Y}$ is a function of $\mathbf{P}_0\mathbf{Y}$, and it gives the chi-squared laws of (b) with noncentralities $\|\mathbf{P}_{\mathcal{T}}\boldsymbol{\theta}\|^2/\sigma^2$ and $\|(\mathbf{I} - \mathbf{M})\boldsymbol{\theta}\|^2/\sigma^2$. The second is zero because $\boldsymbol{\theta} \in \mathcal{C}(\mathbf{Z})$. The vector $\boldsymbol{\theta}$ has entry μ_k in group k , so $\mathbf{M}\boldsymbol{\theta} = \boldsymbol{\theta}$ and $\mathbf{P}_0\boldsymbol{\theta} = \bar{\mu}\mathbf{1}$, and $\mathbf{P}_{\mathcal{T}}\boldsymbol{\theta}$ has entry $\mu_k - \bar{\mu}$ in group k . Its squared length is $\sum_k n_k (\mu_k - \bar{\mu})^2$, which proves Eq. 15.3.3.

For (c): by (a) and (b), F is a ratio of independent chi-squared variables divided by their degrees of freedom, the numerator noncentral, which is the noncentral F law (Definition 4.1.2). Since every $n_k \geq 1$, $\gamma = 0$ iff $\mu_k = \bar{\mu}$ for every k .

For (d), take $\mathbf{X}_0 = \mathbf{1}$ inside $\mathbf{X} = \mathbf{Z}$. Then $\text{SSE}_0 - \text{SSE} = \|(\mathbf{M} - \mathbf{P}_0)\mathbf{Y}\|^2 = \text{SSB}$ by Eq. 6.5.2, with $r - r_0 = g-1$, so F is the statistic of Theorem 11.1.1. In the effects form the hypothesis is $\alpha_k - \alpha_g = 0$, $k < g$: $g-1$ estimable, linearly independent contrasts, so it is testable (Definition 11.3.1). The mean vectors it allows are the constant vectors, so by Theorem 11.3.2(c) its F statistic is the same one. The likelihood ratio property is Theorem 11.4.1, and the power statement is Theorem 11.5.1(a). ■

The noncentrality Eq. 15.3.3 is a weighted sum of squared deviations of the group means from their *weighted* mean. The weights are the group sizes, because a group of n_k observations carries n_k copies of its mean in $\boldsymbol{\theta}$. In a balanced layout $\gamma = m \sum_k (\mu_k - \bar{\mu})^2 / \sigma^2$, the case studied for planning in Proposition 11.5.2 and Example (How many observations per group?). The dependence on the group sizes in unbalanced layouts is taken up in Section 15.5.

15.3.3 Expected mean squares

The distributions in Theorem 15.3.1 need normality. The expectations of the mean squares do not, and they explain why the ratio of two mean squares is the right statistic.

Theorem 15.3.2 — Expected mean squares in the one-way layout

In the one-way model of Definition 15.1.1 (uncorrelated errors with common variance σ^2 , not necessarily normal):

- (a) $E(\text{MSE}) = \sigma^2$;

- (b) $E(\text{MSB}) = \sigma^2 + \frac{1}{g-1} \sum_{k=1}^g n_k (\mu_k - \bar{\mu})^2$;
- (c) in the effects form, $\sum_k n_k (\mu_k - \bar{\mu})^2 = \sum_k n_k (\alpha_k - \bar{\alpha})^2$ with $\bar{\alpha} = n^{-1} \sum_k n_k \alpha_k$, which is $\sum_k n_k \alpha_k^2$ under the side condition $\sum_k n_k \alpha_k = 0$. In a balanced layout it is $m \sum_k (\mu_k - \bar{\mu})^2$ with $\bar{\mu} = g^{-1} \sum_k \mu_k$, which is $m \sum_k \alpha_k^2$ under $\sum_k \alpha_k = 0$;
- (d) $E(\text{MSB}) \geq E(\text{MSE})$, with equality iff $\mu_1 = \cdots = \mu_g$.

Proof. The spaces of Eq. 15.3.1 form an analysis of variance decomposition of $\mathcal{C}(\mathbf{Z})$ in the sense of Theorem 9.6.1, which gives $E(\text{SSE}) = \sigma^2(n-g) + \|(\mathbf{I} - \mathbf{M})\boldsymbol{\theta}\|^2 = \sigma^2(n-g)$ and $E(\text{SSB}) = \sigma^2(g-1) + \|\mathbf{P}_{\mathcal{T}}\boldsymbol{\theta}\|^2$. The squared length was computed in the proof of Theorem 15.3.1. Dividing by the degrees of freedom gives (a) and (b). For (c), $\mu_k = \mu + \alpha_k$ and $\bar{\mu} = \mu + \bar{\alpha}$, so $\mu_k - \bar{\mu} = \alpha_k - \bar{\alpha}$. In a balanced layout $\bar{\mu} = \bar{\mu}$ and $\bar{\alpha} = g^{-1} \sum_k \alpha_k$. (d) follows from (a) and (b). Part (b) is also Exercise (A2), which computes it directly from the group means. ■

Both mean squares estimate σ^2 when the treatments do not differ; otherwise MSB estimates σ^2 plus the treatment variation per degree of freedom. Every F ratio in an analysis of variance follows this logic: the denominator's expectation equals the numerator's exactly when the hypothesis holds (Section 9.6).

The same projections serve the random-effects model of Section 15.1, but the expectations change. With $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I} + \sigma_a^2 \mathbf{Z}\mathbf{Z}^\top$ and a constant mean, $E(\text{MSE}) = \sigma^2$ still, while $E(\text{MSB}) = \sigma^2 + n_0 \sigma_a^2$ with $n_0 = (n - \sum_k n_k^2/n)/(g-1)$, which is m in a balanced layout (Exercise (C1)). The ratio F then tests $\sigma_a^2 = 0$, and the difference of the two mean squares divided by n_0 estimates σ_a^2 . Chapter 32 develops this (Proposition 32.1.1).

15.3.4 The table, derived

Collecting the pieces gives the one-way analysis of variance table. Each row is a subspace, its degrees of freedom are the dimension, its sum of squares is the squared length of the projection of $\mathbf{y}$, and its expected mean square is σ^2 plus the squared length of the projection of $\boldsymbol{\theta}$ per dimension.

Source	Subspace	df	Sum of squares	Mean square	F	Expected mean square
treatments	$\mathcal{T}$	$g-1$	$\sum_k n_k (\bar{y}_k - \bar{y})^2$	MSB	MSB/MSE	$\sigma^2 + \frac{\sum_k n_k (\mu_k - \bar{\mu})^2}{g-1}$
error	$\mathcal{C}(\mathbf{Z})^\perp$	$n-g$	$\sum_{k,j} (y_{kj} - \bar{y}_k)^2$	MSE		σ^2
corrected total	$\mathbf{1}^\perp$	$n-1$	$\sum_{k,j} (y_{kj} - \bar{y})^2$			

Some tables add a row for the mean, with 1 degree of freedom and sum of squares $n\bar{y}^2 = \|\mathbf{P}_0 \mathbf{y}\|^2$, and an uncorrected total $\sum y_{kj}^2$; its F ratio would test $\bar{\mu} = 0$, rarely of interest.

Hand computation used the group totals T_k and grand total T : $\|\mathbf{My}\|^2 = \sum_k T_k^2/n_k$ and $\|\mathbf{P}_0\mathbf{y}\|^2 = T^2/n$, the “correction factor”, so $\text{SSB} = \sum_k T_k^2/n_k - T^2/n$. Such differences of nearly equal numbers lose accuracy in floating point ([Chapter 10](#)); computing deviations first, as the projections do, avoids this.

Example. (The analysis of variance for the nitrogen trial)

For the thirty yields of [Section 15.1](#), with grand mean $\bar{y} = 5.6930$ t/ha:

Source	df	Sum of squares	Mean square	F
nitrogen	4	28.7074	7.1768	26.43
rates				
plots within	25	6.7887	0.2715	
rates				
corrected	29	35.4960		
total				

The mean row would add $n\bar{y}^2 = 972.307$ on one degree of freedom. The critical value is $F_{0.05}(4, 25) = 2.759$, and the p -value is 1.2×10^{-8} . The table says only that the treatment component of the mean is not zero; the contrasts of [Section 15.2](#) and [Section 15.4](#) say what it is. Since the data were simulated, the true noncentrality is known: $\gamma = 103.4$, far more than needed for power near one.

```
import numpy as np
from scipy import stats

rates = np.array([0, 40, 80, 120, 160])          # kg N per hectare
g, m = len(rates), 6                             # 5 rates, 6 plots per rate
true_means = 6.8 - 2.7 * np.exp(-0.018 * rates)  # unknown to the analyst
rng = np.random.default_rng(1515)
group = np.repeat(np.arange(g), m)               # plot i received rate group[i]
y = np.round(true_means[group] + rng.normal(0, 0.5, g * m), 2) # yield, t/ha
n = len(y)
Z = (group[:, None] == np.arange(g)).astype(float) # cell-means model matrix
n_k = Z.sum(axis=0)

M = Z @ np.diag(1 / n_k) @ Z.T
P0 = np.full((n, n), 1 / n)                     # projection onto span(1)
ss_mean = y @ P0 @ y                            # n * ybar^2
ss_between = y @ (M - P0) @ y                   # sum_k n_k (ybar_k - ybar)^2
ss_within = y @ (np.eye(n) - M) @ y             # sum_kj (y_kj - ybar_k)^2
df_between, df_within = g - 1, n - g
F = (ss_between / df_between) / (ss_within / df_within)
p_value = stats.f.sf(F, df_between, df_within)
print(f"{'source':10s} {'df':>3s} {'SS':>9s} {'MS':>8s}")
print(f"{'rates':10s} {'df_between':3d} {'ss_between':9.4f} {'ss_between / df_between':8.4f}")
print(f"{'plots':10s} {'df_within':3d} {'ss_within':9.4f} {'ss_within / df_within':8.4f}")
print(f"{'corrected':10s} {'n - 1':3d} {'ss_between + ss_within':9.4f}")
print(f"F = {F:.2f} on ({df_between}, {df_within}) df, p = {p_value:.2g}")
```

[Run this code online](#)

15.3.5 Three ways to read F

As a proportion of variation. Let $\eta^2 = \text{SSB}/(\text{SSB} + \text{SSE})$, the fraction of the corrected total sum of squares that lies in the treatment space. It is the R^2 of the regression of $\mathbf{y}$ on the group indicators, and the sample version of the correlation ratio of Eq. 14.2.1. By Theorem 9.5.2,

$$F = \frac{\eta^2/(g-1)}{(1-\eta^2)/(n-g)},$$

and in the normal model, under equal means, η^2 has the Beta law of that theorem, with $E(\eta^2) = (g-1)/(n-1)$, not small with many groups and few observations. In the nitrogen trial $\eta^2 = 0.809$: four fifths of the variation among plots is variation among rates.

As the largest contrast. By Proposition 15.2.1(d), every contrast has $\text{SS}(\mathbf{c}) \leq \text{SSB}$, with equality for $c_k \propto n_k(\bar{y}_k - \bar{y})$. Dividing by s^2 , the largest squared t statistic over all contrasts is $(g-1)F$. Hence the F test rejects iff some contrast, perhaps not an interesting one, is declared nonzero by Scheffé's method (Corollary 13.2.3).

As a two-sample test, for $g = 2$. Then $\mathcal{T}$ is a line, the only contrast is $\mu_1 - \mu_2$ up to scale, and $F = T^2$ for the pooled two-sample t statistic of Exercise (B2), whose sensitivity to unequal variances the one-way test inherits (Section 15.5).

15.3.6 Exercises

15.3.6.1 A. Check your understanding

Exercise A1.

A one-way experiment with four treatments and six observations per treatment gives a treatment mean square of 12.0 and a corrected total sum of squares of 96.0. Complete the analysis of variance table, compute F and η^2 , and state the reference distribution of F under equal means.

Solution. The degrees of freedom are 3, 20 and 23. The treatment sum of squares is $3 \times 12 = 36$, so the error sum of squares is $96 - 36 = 60$ and $\text{MSE} = 3$. Then $F = 12/3 = 4$, $\eta^2 = 36/96 = 0.375$, and under equal means $F \sim F(3, 20)$.

Exercise A2.

Show from Eq. 15.3.2 that $\text{SSB} = \sum_k T_k^2/n_k - T^2/n$ and $\text{SSE} = \sum_{k,j} y_{kj}^2 - \sum_k T_k^2/n_k$.

15.3.6.2 B. Practice

Exercise B1.

For $g = 2$ with $n_1 + n_2 = n$ fixed, show that $\gamma = (\mu_1 - \mu_2)^2 n_1 n_2 / (n\sigma^2)$, and that it is largest for $n_1 = n_2$. By how much does the noncentrality fall for the split $n_1 = n/4$, $n_2 = 3n/4$?

Solution. With $\bar{\mu} = (n_1\mu_1 + n_2\mu_2)/n$, $\mu_1 - \bar{\mu} = n_2(\mu_1 - \mu_2)/n$ and $\mu_2 - \bar{\mu} = -n_1(\mu_1 - \mu_2)/n$. So $\sigma^2\gamma = (\mu_1 - \mu_2)^2(n_1n_2^2 + n_2n_1^2)/n^2 = (\mu_1 - \mu_2)^2n_1n_2/n$. The product n_1n_2 with fixed sum is largest at $n_1 = n_2 = n/2$, where it is $n^2/4$. For the split $(n/4, 3n/4)$ it is $3n^2/16$, so the noncentrality falls by a quarter.

Exercise B2.

In the normal model with equal means $E(\eta^2) = (g-1)/(n-1)$, which is not zero, so η^2 overstates the treatment variation in small experiments. Show that the numerator of

$$\hat{\omega}^2 = \frac{\text{SSB} - (g-1)\text{MSE}}{\text{SSB} + \text{SSE}}$$

has expectation $\sum_k n_k(\mu_k - \bar{\mu})^2$, whatever the means, so that it removes the bias from the numerator. When is $\hat{\omega}^2$ negative?

Solution. By Theorem 15.3.2, $E(\text{SSB}) = (g-1)\sigma^2 + \sum_k n_k(\mu_k - \bar{\mu})^2$ and $E\{(g-1)\text{MSE}\} = (g-1)\sigma^2$, so the difference has the stated expectation. The numerator is negative iff $\text{MSB} < \text{MSE}$, that is, iff $F < 1$, which often has null probability above one half: 0.57 for the nitrogen design's $F(4, 25)$ in the normal model. The ratio itself is not unbiased, since its denominator is random.

Exercise B3.

In the nitrogen trial, compute the Scheffé-maximal contrast $c_k = n_k(\bar{y}_k - \bar{y})$ from the rate means, and verify that its squared t statistic is $4F$. Describe in words what it compares.

15.3.6.3 C. Going deeper

Exercise C1.

In the random-effects model $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I} + \sigma_a^2\mathbf{Z}\mathbf{Z}^\top$, $E(\mathbf{Y}) = \mu\mathbf{1}$, use Theorem 2.3.1 to show that $E(\text{SSB}) = (g-1)\sigma^2 + (n - \sum_k n_k^2/n)\sigma_a^2$ and $E(\text{SSE}) = (n-g)\sigma^2$. Show that $n_0 = (n - \sum_k n_k^2/n)/(g-1) \leq n/g$, with equality iff the layout is balanced.

Solution. By Theorem 2.3.1, $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \text{tr}(\mathbf{A}\Sigma) + \boldsymbol{\theta}^\top \mathbf{A} \boldsymbol{\theta}$, and $\boldsymbol{\theta} = \mu\mathbf{1}$ is annihilated by both $\mathbf{P}_{\mathcal{T}}$ and $\mathbf{I} - \mathbf{M}$. Now $\text{tr}(\mathbf{P}_{\mathcal{T}}) = g-1$ and $\text{tr}(\mathbf{P}_{\mathcal{T}}\mathbf{Z}\mathbf{Z}^\top) = \text{tr}(\mathbf{Z}^\top \mathbf{M} \mathbf{Z}) - \text{tr}(\mathbf{Z}^\top \mathbf{P}_0 \mathbf{Z})$. Since $\mathbf{M}\mathbf{Z} = \mathbf{Z}$, the first term is $\text{tr}(\mathbf{Z}^\top \mathbf{Z}) = n$. The second is $n^{-1}\|\mathbf{Z}^\top \mathbf{1}\|^2 = \sum_k n_k^2/n$. For the error, $(\mathbf{I} - \mathbf{M})\mathbf{Z} = \mathbf{0}$, so only $\sigma^2 \text{tr}(\mathbf{I} - \mathbf{M}) = (n-g)\sigma^2$ remains. Finally $\sum_k n_k^2 \geq n^2/g$ by the Cauchy-Schwarz inequality, with equality iff all $n_k = n/g$, so $(g-1)n_0 \leq n - n/g = (g-1)n/g$.

Exercise C2.

Show that F is unchanged when $\mathbf{y}$ is replaced by $a\mathbf{y} + b\mathbf{1}$ with $a \neq 0$, and when the observations are permuted within groups. Show that it changes when a constant is added to the observations of one group only, and explain why it should.